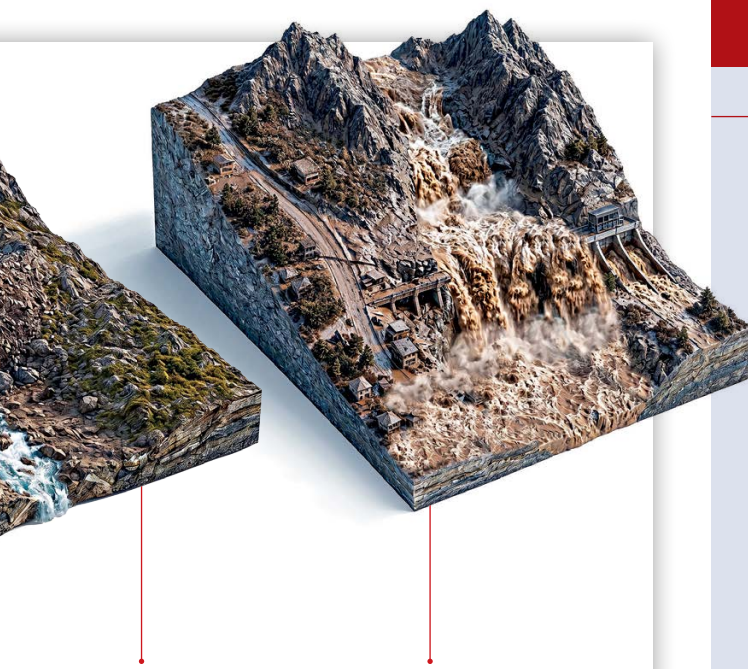


THE LAKE BECOMES UNSTABLE
The dam may breach because of a landslide, avalanche, falling glacier mass or earthquake

A DESTRUCTIVE FLOOD RACES DOWNSTREAM
A torrent of water mixed with ice, rocks and sediment surges past narrow valleys, destroying roads, bridges, hydel projects and settlements. The flood can travel many kilometres before losing force

WHY INDIA IS VULNERABLE

The country is already within the glacial floods danger zone. It has a dynamically changing high-altitude hazard system stretching some 2,500 km from Ladakh to Arunachal

- Indian agencies have mapped roughly 7,500 glacial lakes within India, out of around 28,000 across the wider Himalayan region
- Himalayan temperatures have risen by roughly 0.15°C to 0.60°C over the past decade
- Snow surface temperatures on the Indian side have increased from –11.27°C in 2000–09 to –7.13°C in 2020–23
- In 2024, ISRO examined 2,431 Himalayan glacial lakes larger than 10 hectares and found that 676 had expanded substantially between 1984 and 2023
- Worryingly, 601 of the 676 had more than doubled in area. Of them, 130 lay within India
- A 2025 assessment led by the National Disaster Management Authority examined 189 high-risk glacial lakes, of which 56 have now been classified as 'very high risk'
- During the 2025 monsoon, ISRO monitored 2,024 glacial lakes of one hectare or more lying in the catchments of just 26 NHPC hydropower projects



"Whatever happened in Nepal can happen in India. Political boundaries really don't exist from a mountain-stability perspective"

DR UMESH HARITASHYA
Professor, Earth, Environment & Sustainability, University of Dayton, Ohio

tion (ISRO), 16,627 of these fall within India's political boundaries while 34,919 overall feed the Indus, Ganga and Brahmaputra basins, on which India depends. It maps about 7,570 glacial lakes larger than 0.25 hectares within India and more than 28,000 across the wider Himalayan region.

These are not static blue dots. In a long-term satellite assessment in 2024, ISRO examined 2,431 lakes larger than 10 ha across the Himalayan river basins and found that 676 had expanded significantly since 1984. Of them, 130 lay within India: 65 in the Indus basin, seven in the Ganga and 58 in the Brahmaputra basins. An extraordinary 601 had more than doubled in area. In fact, the Ghepang Ghat lake in Himachal had expanded by 178 per cent between 1989 and 2022, from 36.5 ha to 101.3 ha.

Bigger lakes hold more water, but size alone does not determine danger. A smaller lake perched above a steep, inhabited gorge may be deadlier than a larger, stable one. The strength of its moraine wall, the possibility of an avalanche entering it, the gradient and shape of the valley, and the people and infrastructure downstream all matter. That is why experts say a simple national count of 'high-risk lakes' can produce false reassurance. India must assess whole catchments—lakes, glaciers, rock faces, permafrost, river blockages and downstream exposure—rather than watch only a list of water bodies.